

CENG0075 MSc Research Project

Anomalous Voltage Responses Mechanisms Preceding Thermal Runaway in Cylindrical Lithium-Ion Batteries

by: Tianli Shi

Student's full Name and SURNAME (surname in capitals)

Student's registration number: TSHIX72

Supervisor(s): Alex Rettie

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Department of Chemical Engineering

University College London (UCL)

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Declaration

I, Tianli Shi, confirm that the work presented in this thesis is my own.

Where information has been derived from other sources, I confirm that this has been indicated in the thesis.

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.....
Tianli Shi
Student's Name

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Tianli Shi
Student's Signature

.....
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Abstract

The lithium-ion battery usually has electrical, thermal, structural, and material state changes before thermal runaway. The mechanism of early anomalous terminal voltage responses is not fully explained. This research uses the 100% SOC B18650CX-20M as the sample and analyzed its voltage change and material change processes before thermal runaway. It uses the ARC to detect the voltage, temperature, and self-heating rate and cooperates with the X-ray CT, TGA, and DSC to do the characterization. There are two discrete voltage drop events before the rapid temperature rise. During the drop 1, the terminal voltage decreased from approximately 3.98 V to 2.45–2.50 V. During the drop 2, the voltage fell from approximately 2.25 V to 0.20 V and subsequently recovered to 3.4–3.5 V. This is different from the simple continuous discharging process and permanent internal short circuit. The material analysis shows the cathode is made of mixed LMFP/layered-oxide. The research shows that the low temperature side reactions, volatilization, pressure development, transient electrical-contact changes, and possible activation of the current-interrupt device might contribute to the anomalous voltage responses. Though the current results can not distinguish safety-device activation from localized internal short circuits, the observed voltage drop and recovery can be the early warning signal for escalating failure. This research supports the development of multi-signal warning strategies integrating voltage, surface temperature, and self-heating rate. The replicated and in-situ experiments are needed to prove the repeatability and mechanistic specificity of these signals in the future.

Contents

1	Introduction	3
1.1	Background	3
1.2	Ethical Statement	4
1.3	Problem Statement	4
1.4	Objective	5
2	Literature Review	7
2.1	Industrial, Academic and Safety	7
2.1.1	Growth of Li-ion battery use in Net Zero systems	7
2.2	Li-ion Cell Structure and Failure Pathways	8
2.2.1	Basic cell components	8
2.2.2	Failure progression and critical review	8
2.3	Voltage Activities	10
2.4	Experimental and Simulation Techniques for Failure and Emission Measurement	11
2.4.1	Experimental Diagnostics: X-ray CT and XRD	11
2.4.2	Experimental Diagnostics: ARC, TGA and DSC	11
3	Methodology	13
3.1	Cell preparation and baseline characterization	13
3.2	ARC testing and identification of voltage events	14
3.3	XRD Analysis	15
3.4	DSC Analysis	16
3.4.1	DSC temperature programme	16
3.4.2	DSC parameter extraction	16
3.5	TGA Analysis	17
4	Results and Discussion	18

4.1	CT Results	18
4.2	XRD Results	19
4.3	DSC Results	22
4.4	TGA Results	25
4.5	Thermal response and voltage evolution in ARC Test	27
5	Conclusion	31
5.1	Battery structure and material state prior to failure	31
5.2	Abnormal Voltage Response Prior to Thermal Runaway and Its Mechanistic Implications	32
5.2.1	Drop 1 Conclusion	32
5.2.2	Drop 2 Conclusion	32
5.3	Limitations	33
5.4	Future suggestions	34

Chapter 1

Introduction

1.1. Background

As global actions are executed to handle climate change, push the energy transformation and pursue of “Net Zero” in depth, transportation electrification and large-scale energy storage systems are developing more and more faster. During this evolution, Li-ion batteries become the key driving force due to their high energy density, long cycle life, and technology maturity. However, with the growth of the installed base and application scenarios, the safety issues of it are gradually becoming a bottleneck limiting the development of this technology.

The failure of Li-ion batteries is usually caused by mechanical, electrical, or thermal abuse. These harsh external excitations will damage the internal separator of batteries, thereby causing the batteries to enter an irreversible state of “thermal runaway”. During the thermal runaway process, the internal stress of batteries increases instantly, and then a large amount of gaseous products and solid particles are released. These emissions are not only flammable, but they are easily able to lead to fire and even explosions. They contain some toxic and harmful components.

Overall, in-depth exploration of failure modes and emission characteristics of lithium-ion batteries under abuse conditions has great academic value and practical significance. This experiment aims to use high-throughput methods to systematically classify and quantify the gas and particulate matter generation patterns of lithium-ion batteries under different abuse environments. This not only offers data support for safer battery system designs (such as explosion-proof valve improvement and thermal isolation design), but also reveals the electrochemical and physical connections between batteries from microscopic degradation to macroscopic failure. It lays a

solid foundation for the next generation of safer and more resilient energy storage solutions [1].

1.2. Ethical Statement

This research aims to explain the anomalous voltage responses and material degradation mechanism, thereby supporting reference for battery fault identification, thermal runaway warning, and safety design. This study does not involve human participants, animal experiments, or personal data. The thermal runaway, cell open, and material analysis can create engineering ethics and safety issues such as flammable and toxic gases, ejected materials, and hazardous chemical waste. The relevant experiments should be conducted only after the risk assessment has been completed and after following the approved safety operating procedures. Suitable containment, ventilation, remote monitoring, personal protection, and emergency measures, thereby reducing the potential risks to researchers, facilities, and the environment. Regarding the responsibilities of environment and resources, this research limits the number of batteries while ensuring reliable results and enough replication. Simulation and non-destructive technology are applied to reduce unnecessary destructive testing. Failure cells, electrolyte liquid, and electrode materials are collected and handled with hazardous waste management procedures. In data analysis and reporting, the research will save and faithfully present original results, anomalies, uncertainties, and limitations of the methodology. The explanation of the mechanism is proposed by combining the results of electrical, thermal, structural, and material analysis. All the conclusions are limited to the cell system and experimental conditions in this study.

1.3. Problem Statement

The lithium-ion battery is widely needed due to the Net Zero battery scaling up and global energy systems electrification. Though these energy storage system are important to the decarbonization process, their widespread application has raised safety concerns, especially the battery failure problems related to degradation. In order to reduce these risks and improve the battery management system, deeply understanding the battery failure modes and emission characteristics is important. Current literature has done a lot of research about X-ray CT, ARC, TGA, DSC, and XRD analysis. But these insights are a little fragmented. This report aims to sort out the theoretical bases for commercial lithium-ion battery characterization in different

failure modes. This report combines internal failure mechanisms, electrochemical diagnostic symptoms, structural evidence following failure, and final emissions. Several key questions are analyzed to understand current progress.

- The first is the main failure mode of commercial Li-ion batteries in abuse conditions. It also includes the range of gases and particles along with state of charge, electrochemical system and different types of abuse.
- The second is about the characterization technologies. How the mechanisms of ARC, TGA, DSC, and XRD indicate the internal failure process and severity after combining destructive post-mortem opening.
- The third is the methods to design high-throughput and reproducibility in order to construct the data sheet that connects battery failure characteristics and emission profile.

1.4. Objective

Assumptions

The following assumptions are the basis of this study. This research assumes the selected cylindrical lithium-ion battery can represent the cells that have a similar chemical system and structural design. Meanwhile, it assumes its electrode voltage can show the internal electrochemical state and material degradation process. The recorded anomalous voltage response is thought mainly to be from internal impedance, polarization state, active lithium density, and active material integrity. This research assume failure process has time continuity and repeatability. The material characterization results are assumed to reflect the material condition at the end of the test. But consider the non-uniformity in cylindrical battery, the localized sampling results is not thought to be the same state in all the region of cell. All the conclusions are based on the battery types, state of charge, and failure conditions specified in this study.

General Objective

This research aims to reveal the formation mechanism of anomalous voltage responses before thermal runaway in a cylindrical lithium-ion battery. It focus on construct the relation between voltage characteristics and the degradation processes of main materials of electrodes, separators, electrolytes, and current collectors, thereby understand the battery failure evolution path and evaluate the potential of anomalous voltage signal as early warning indicators for thermal

runaway.

Specific Objective

- Objective 1: The system can identify the voltage drop, recovery, fluctuation, or transient rebound and determine their time order relative to the temperature rise and thermal runaway.
- Objective 2: Find out how processes such as structural changes of electrodes, SEI layer decomposition, and separator melting change the internal state of the battery.
- Objective 3: Explain how the loss of active lithium, loss of active material, localized short circuit, and electrode polarization transform the materials changes into an anomalous voltage signal.
- Objective 4: Establish a mechanistic workflow that shows micro-scale material changes to macro-scale and temperature responses. Identify the failure stages and the transition points.
- Objective 5: Evaluate whether the voltage has enough lead time, repeatability, and mechanistic specificity, so that it can be used in the early detection of thermal runaway in a cylindrical lithium-ion battery.

Chapter 2

Literature Review

2.1. Industrial, Academic and Safety

2.1.1. Growth of Li-ion battery use in Net Zero systems

As the key support of global decarbonization and the Net Zero goal, lithium-ion batteries are experiencing explosive growth. Electrical vehicles are replacing traditional ICE vehicles and ESS and power grid peak-shaving equipment are deploying lithium-ion batteries. The comprehensive and massive-scale application makes the safety performance gradually become a macro-level security issue [2]. When thermal runaway or other failures occur, the risks are far more than temperature rise and localized combustion, a more severe challenge is the associated complex emissions. The first is toxic and flammable gases such as carbon monoxide, hydrogen, and various volatile organic compounds. These gas not only has the risk of triggering a secondary explosion but can also cause acute poisoning. The second is high-risk chemical exposure such as Electrolyte salts (e.g., LiPF_6). Its decomposition at high temperature produces highly toxic and corrosive hydrogen fluoride gas as well as fluorides such as phosphoryl fluoride [3]. The third is smoke and particles. The degradation process releases large quantities of aerosols, black carbon, and fine particulate matter containing toxic heavy metals (such as nickel, cobalt, and manganese). These emissions have high accumulating risks, the gas and smoke are difficult to disperse in a short period of time [3].

2.2. Li-ion Cell Structure and Failure Pathways

2.2.1. Basic cell components

The anode is usually made of layered graphite materials, it is coated onto the copper foil current collector. The anode is in charge of lithium intercalation during charging. The cathode mainly uses the layered transition metal oxides such as high-nickel NCM/NCA or high thermal stability lithium iron phosphate. It is coated on an aluminium foil current collector. The membrane is a porous polyolefin film (such as PE, PP, or a three-layer composite membrane). It is used to isolate the positive and negative electrodes and allow lithium ions to pass through. The electrolyte is made of non-aqueous carbonate solvents (such as EC, DMC, EMC) and lithium hexafluorophosphate (LiPF_6) solute. The packaging and security architecture are made of a robust steel or aluminium shell with an internal jelly-roll structure and a top cover integrating a current cut-off device and a pressure relief valve [4].

2.2.2. Failure progression and critical review

From the onset of abuse to the eventual occurrence of catastrophic thermal runaway, the typical chain development path is as follows:

Different research has different views on the trigger of temperature and the dominant mechanisms at key nodes. Traditional research thought SEI starts exothermic decomposition when the temperature is between 80 °C to 120 °C. But the newest thermal analysis shows the side reactions between lithiated graphite and the electrolyte at the negative electrode can be detected at 60 °C in a commercial high-density battery. This directly affects the capture of early warning signals from the battery [5] [6].

Some researchers proposed that separator melting triggers widespread internal short circuits, thereby driving the reaction to irreversibility. Another group of researchers thought the localised internal short circuit happens when the separator does not reach its melting point under mechanical or electrical abuse, such as lithium plating. This kind of hotspots will cause localized vaporization of the electrolyte, thereby causing the safety valve to blow open prematurely at a lower temperature. For high-nickel NCM batteries, the phase transition of the cathode material occurs between 180 °C to 220 °C and releases active oxygen, this is often seen as the key part of triggering severe thermal runaway. However, the research shows the crystal structure of the LFP

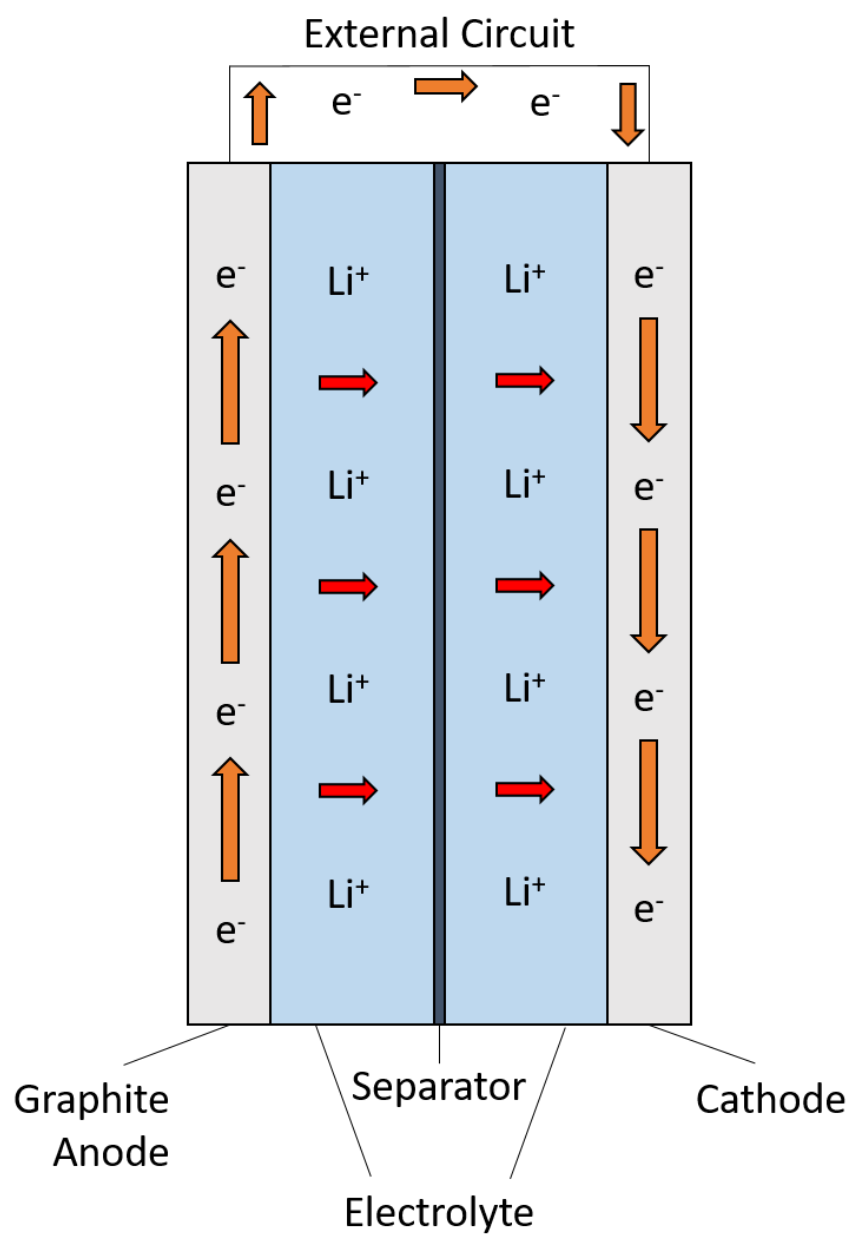


Figure 2.1: Li-ion Cell Structure

system is stable, with nearly no oxygen released. Its thermal runaway is driven by continuous exothermic side reactions between the negative electrode and the electrolyte, as well as solvent vaporization [7].

Overall, the evolution of failure path are not only influenced by abuse conditions, but it is also deeply connected to the chemical system and structural design. This directly causes the venting time and emission severity to vary in different experimental conditions, thereby posing challenges for the creation of a unified dataset.

2.3. Voltage Activities

Some researches has shown the cylindrical lithium-ion battery might perform multi stage voltage changes during thermal runaway. This include non-zero voltage drop in early stages, voltage drop near the pressure relief point, transient recovery voltage, and final voltage drop to zero. The results of some research shows this phenomenon might relate to temporary changes in top cover, positive tab, or devices of current changes. The internal short circuit can undergo before thermal runaway, while the occurrence does not mean battery will enter uncontrolled temperature rise stage [7].

The current interrupt devices (CID) is set at the top part of the battery. When the side reaction producing gases, increase of the internal pressure might trigger CID or makes deformation of top cover devices, thereby disconnecting the connection between cathode and external terminal. Researchers found that CID might trigger before the external venting is observed. It shows voltage drop, internal pressure change, and pressure release needs to be defined separately [8]. The voltage recovery after the pressure release is a special response in thermal runaway. This does not means the battery back to normal state. However, the internal short circuit pathway or the connection status between external measurement terminal and internal cell core undergoes transient changes. Researchers gives a possible mechanism to explain this phenomenon. After the pressure release, as the internal pressure reducing and structural displacement occurring, part of the electrical connection might re-established, thereby makes electrode voltage recovery. Conventional pressure-triggered CIDs are designed to disconnect irreversibly, thereby the more reasons such as tab connection, floating measurement terminal should be considered to explain voltage recovery [9].

2.4. Experimental and Simulation Techniques for Failure and Emission Measurement

2.4.1. Experimental Diagnostics: X-ray CT and XRD

- X-ray CT offers the non-destructive 3D spatial imaging, especially suitable for commercial jelly-roll structure of cylindrical battery. By comparing 3D tomographic images before and after failure, the casing deformation, safety valve in the blown-open electrode delamination can be observed. This supports the explanation of “how specific mechanical failure modes trigger discharge of specific intensities”.
- The XRD are applied to test the crystal structure and phase evolution of cathode and anode materials. Through the comparison of diffraction patterns of the residual solid phases after thermal runaway, XRD can directly prove whether the cathode material has phase transition from a layered structure to a spinel or rock-salt phase. Meanwhile, through the analysis of XRD pattern of the anode material, the residual state of lithium in graphite interlayers are shown clearly, thereby supporting the evidence for heat release and gas generation from the reaction between the negative electrode and the electrolyte.

2.4.2. Experimental Diagnostics: ARC, TGA and DSC

- The Accelerated Rate Calorimetry (ARC) is the core device for testing the thermal runaway behavior of commercial cylindrical batteries. In the adiabatic environment, the ARC can accurately capture the milestones of the battery behaviour from self-heating onset temperature, venting temperature, to thermal runaway onset temperature. Through recording the temperature rise rate under adiabatic conditions and pressure changes, ARC supports the data for hazard classifications of batteries with different states of charge (SOC) and chemistries.
- The Thermogravimetric Analysis (TGA) is used to measure the mass loss of a sample during heating. By monitoring the mass loss curves under different atmospheres, the TGA can reflect the volatilization or thermal decomposition temperature of the electrolyte solvent, the proportion of mass loss due to SEI decomposition and, mass change triggered by oxygen release from the cathode [10].

- Differential Scanning Calorimetry (DSC) is also used at the material level. It is used for measurement of heat absorption and release by electrode materials and electrolytes during heating. The DSC can detect the microscopic thermodynamic peaks associated with SEI layer decomposition, separator melting, reaction between anode and electrolyte, and the structural phase transition of the cathode. Combining thermodynamic methods, DSC can be used to calculate the activation energy of each step. This provides theoretical support for thermochemical reaction pathways of emission products [11].

Chapter 3

Methodology

3.1. Cell preparation and baseline characterization

The experiment uses the B18650CX-20M cylindrical lithium-ion cell which is made by BAK battery company. According to the manufacturer's specifications, basic details are listed in the sheet.

Parameter	Specification
Cell type	Cylindrical rechargeable lithium-ion cell
Model	B18650CX-20M
State of charge	100% SOC
Rated capacity	2,000 mAh
Typical capacity	2,050 mAh
Nominal voltage	3.6 V
Rated energy	7.20 Wh
Typical energy	7.38 Wh
Cell mass	≤ 45 g

Table 3.1: Key specifications of the lithium-ion cell.

Based on the safety data sheet for lithium-ion cell materials provided by the manufacturer, the materials are listed in the sheet. Before the ARC test, all the cells are assigned a number. The visual inspection includes checking heat-shrink film, leakage, corrosion, dents, top deformation, or abnormal initial voltage. Only batteries in the permissible range can be used for later

Component	Chemical description	Approx. mass fraction
Lithium transition-metal oxide	$\text{Li}(x)\text{MO}_2$; M may include Co, Ni, Mn, or Al	Approx. 37.3%
Carbon (graphite)	C	Approx. 21.0%
Aluminum	Al	Approx. 3.27%
Copper	Cu	Approx. 7.69%
Steel can	Predominantly Fe	Approx. 13.53%
Electrolyte, total	EC, EMC, DMC, and LiPF_6	Approx. 10.67%
Other proprietary materials	May include separator, binders, and other auxiliaries	Approx. 6.54%

Table 3.2: Chemical composition and approximate mass fraction of battery components

experiments, thereby reducing the impact of initial state differences on the final results. In order to set the structural standards, X-ray computed tomography is applied before the ARC test. The X-ray CT focuses on recording the jelly roll alignment, central cavity, relative positions of electrode sheets, top cap and tab areas, thereby further ensuring the integrity of the samples. Battery cells may rupture, catch fire and leak internal materials, all ARC tests are conducted in a controlled environment with explosion-proof capabilities, local exhaust systems, and gas treatment facilities.

3.2. ARC testing and identification of voltage events

- After the charging process, set the 100% SOC cell in the test chamber of the accelerating rate calorimeter. The anode and cathode are connected to voltage acquisition channels through high-temperature resistant insulated wires. It is an open-circuit state and voltage is monitored. The thermocouple is attached to the wall at the axial point of the cell and secured with tape. This step can reduce thermal contact error and displacement.
- The ARC test uses a heat-wait-search (HWS) programme. The test starts from 50 °C and 5 °C step size. Every time the temperature reaches the set value, it will wait for 30 minutes, thereby making the cell and chamber reach thermal equilibrium. Then it enters 10-minute self-heating search phase. When the rate of temperature rise of the cell

reaches or exceeds $0.02\text{ }^{\circ}\text{C min}^{-1}$ during the search phase, the system will switch to adiabatic tracking mode, this allows the temperature of the chamber to change with the cell. Meanwhile, the self-heating process is continuously recorded. The test continues until thermal runaway occurs or reaches the safety termination conditions of the equipment.

- The voltages are recorded based on the synchronously recorded terminal voltage-time curves. The baseline drift and noise range of the channel are determined by a stable voltage interval before the onset of self-heating. When the voltage change exceeds the noise range and lasts for at least three sampling points, the event is set as valid [12]. This method can eliminate the fake signals from the single-point pulses and acquisition switching.
- The original voltage curve and first-order time derivative dV/dt are used for determining the location of the event. The characteristic moment of an event is defined as the time when dV/dt reaches the minimum value in the interval. Based on the time and waveform of the voltage signal, the effective events are marked as initial decline (V_{E1}), voltage drop-and-recovery event (V_{E2}), and the voltage drops irreversibly to the instrument's lower limit (V_{E3}). If one of the waveforms does not appear during the test, then no assignment is performed [13].

3.3. XRD Analysis

Apply X-ray diffraction to characterize the phase composition and crystal structure of electrode materials. XRD is used to establish the correlation between voltage-temperature events and the crystal structure status of electrodes during ARC testing. The anode and cathode samples are taken from fixed positions on the jelly roll, such as the axial middle part of the coil core and near the thermocouple location. During the XRD test, the samples are kept rotating. A Cu-target X-ray source is used, the generator voltage and tube current are set to 45kV and 40mA. The wavelengths of Cu $K\alpha_1$ and $K\alpha_2$ radiation are $1.540598\text{ \AA}$ and $1.544426\text{ \AA}$. The irradiation length is set to 5.0mm, and the sample length is set to 10.0mm. The scan time recorded in the instrument documentation is 400.05s. The actual angular range from the output data is approximately $5.0033 - 80.0035^{\circ}$. The median angular interval is 0.033423° . Initial peak searching, phase matching, and profile fitting are performed using PANalytical HighScore software. The peak searching is applied using the minimum second-derivative method and phase identification

uses a multi-phase search mode. The diffraction peak positions of phases are calculated based on crystal structure data from the Crystallography Open Database. The main references include LMFP (COD 1524250), layered NMC811 (COD 1520789), and metallic Al (COD 9008460). For diffraction peaks that are close or overlap, multiple candidate phases can be marked at the same time. The peak coverage is only used for candidate phase selection, it does not represent the phase mass fraction [13].

3.4. DSC Analysis

The LMFP/NMC cathode and anode of the coin cell tested in the DSC are the same with cylindrical battery. The test cell does not have a cathode.

3.4.1. DSC temperature programme

The temperature of the DSC is made of five continuous stages. Firstly, heat the sample temperature from 23 °C to 105 °C at a rate of 2.0 K min⁻¹. Then maintain it at a constant temperature of 105 °C for 59 minutes. The third step is to heat from 105 °C to 110 °C at 2.0 K min⁻¹. Then maintain a constant temperature of 110 °C for 59 minutes. Last step is cool from 110 °C to 25 °C at 0.5 K min⁻¹.

3.4.2. DSC parameter extraction

- The first is the exothermic onset temperature. Apply the tangent intersection method, which means fitting the pre-reaction baseline and tangent to the rising part of the heat flow curve. Then the temperature at the intersection of the two lines is set as exothermic onset temperature.
- The second is peak temperature. The measured temperature matched to the point where the fixed heat flow reaches a peak value in the reaction range is set as apparent peak temperature.
- The last is apparent heat of reaction. In the set time of t_1 and t_2 , integrate the differential heat flow after removing the local baseline.

$$\Delta H_{\text{app}} = \frac{1}{m} \int_{t_1}^{t_2} [\Delta \dot{Q}_{\text{meas}}(t) - \dot{Q}_{\text{baseline}}(t)] dt, \quad (3.1)$$

3.5. TGA Analysis

In order to analyze the mass changes of cathode materials, thermogravimetric analysis is applied. After opening the battery cell, scrape the cathode surface to get coating powder, it will serve as the test sample. The temperature program covers the range of 25 – 110 °C, including an isothermal stage of 105 °C. This kind of setup can monitor the volatilization behavior of absorbed water, residual solvents, and components related to the electrolyte. Continuously record the change of sample mass with temperature and time. When handling the data, set the sample mass at m_0 at 25 °C as the reference, the measured mass m_t at different temperatures and times is normalized to the residual mass percentage [14].

$$w_t = \frac{m_t}{m_0} \times 100\% \quad (3.2)$$

The mass loss rate is calculated based on the following formula:

$$\Delta w = \frac{m_0 - m_t}{m_0} \times 100\% \quad (3.3)$$

Record the mass at the start and end points of the heat stage and isothermal stage. Calculate the mass change during every stages.

Chapter 4

Results and Discussion

4.1. CT Results

The CT images clearly show key structural components such as positive terminal cap, crimped casing edge, wound core, and the steel housing. The jelly roll is formed by winding the positive electrode, separator, and negative electrode, it appears as a parallel layered structure. The coil core remained intact, no overall collapse or casing deformation is observed.

The positive electrode end-cap region has a distinct multi-layered structure. From the external part to the internal part, the positive terminal cap, top cover plate, pressure relief disk or current interrupt device (CID), and positive collector tab can be identified. The end cap assembly is secured to the top of housing through rolled-edge structure. The insulating gasket is located between the end cap and the steel shell. This design can maintain electrical insulation and sealing. The positive tab connects the positive current collector of the jelly roll and the top cap, thereby forming a conductive path from the interior of the cell to the external positive terminal. The pressure relief disk is set at the positive electrode path. When the internal pressure increases to the design threshold, this structure can deform and trigger the CID, thereby breaking the internal path [15].

This CT result provides a vital initial structural reference for subsequent analysis of voltage responses. The CID, positive tab and end cap are concentrated in the top region. The resistance change due to the internal pressure, thermal expansion, or local structural deformation might lead to a terminal voltage drop or electrical disconnection.

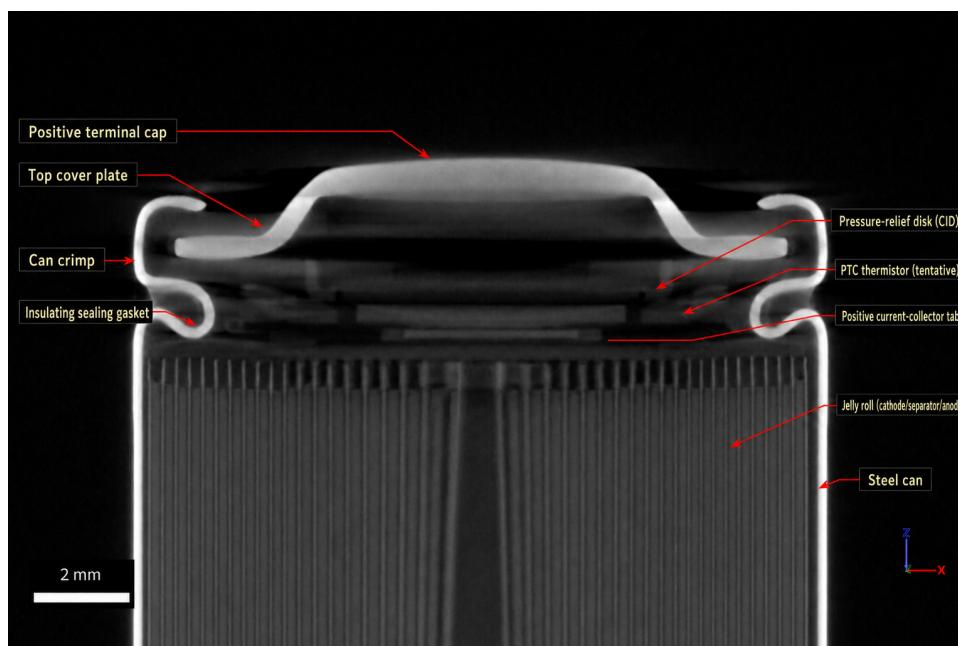


Figure 4.1: Axial CT cross-section of positive electrode of normal cylindrical lithium-ion battery

4.2. XRD Results

After the correcting for scan time, the major diffraction peaks obtained from the rapid scan and long-duration scan show excellent consistency in position and relative intensity. The correlation coefficient reached $r=0.9941$ in the range of $5^\circ \leq 2\theta \leq 50^\circ$. The angular offset is nearly -0.000° . This shows these two scanning modes did not generate a detectable systematic peak shift. The main diffraction characteristics have good repeatability [16]. The long-duration scan shows smoother signals and clearer weak peaks, thereby used for later correction and phase screening.

The long-term scan curves exhibit increasing signals with increasing θ . This signal might be influenced by various reasons, such as sample preparation, amorphous carbon and binders, and fluorescence effects. It can not be interpreted as the amorphization of the cathode material. This study uses the asymmetric least squares method to perform the fit and subtraction, the main sharp diffraction peaks are preserved, and the recognizability of low-intensity peaks is improved.

The strongest peak after correction is located at nearly 35.6° , the other peaks are distributed in the region of $17-21^\circ$, $25-30^\circ$, $36-45^\circ$, and above 50° . The cathode contains crystalline components with high crystallinity.

After the initial matching of experimental peak positions and the reference peaks, most main diffraction peaks match the reference diffraction positions of olivine-type LMFP. Meanwhile, some experimental peaks match the reference peaks of layered transition metal oxides, especially at $36\text{--}45^\circ$, $58\text{--}60^\circ$, $64\text{--}70^\circ$, $77\text{--}78^\circ$. These results support the possibility that the cathode contained both LMFP and layered oxide components before failure.

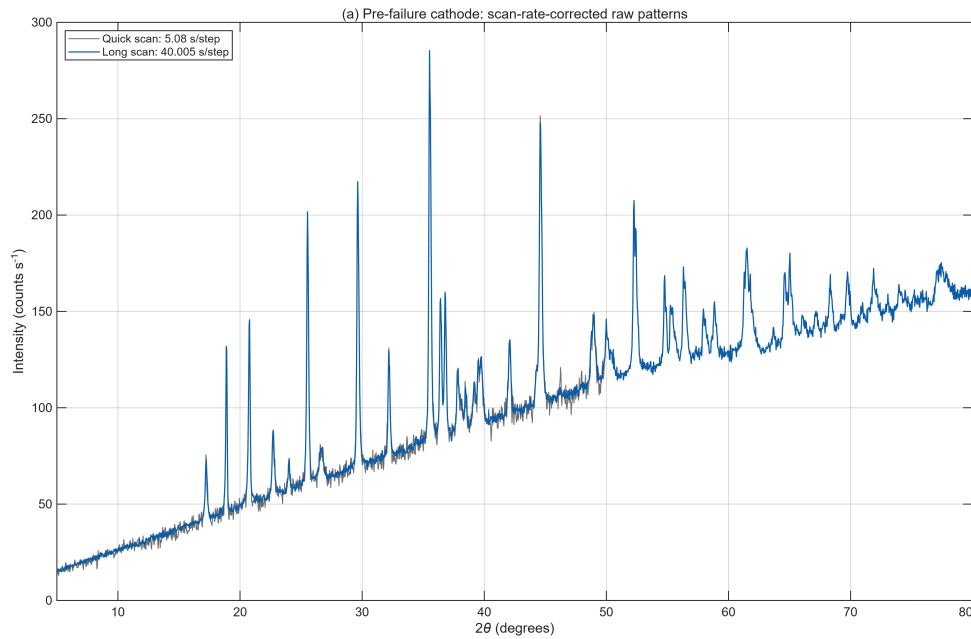


Figure 4.2: Raw diffraction curve

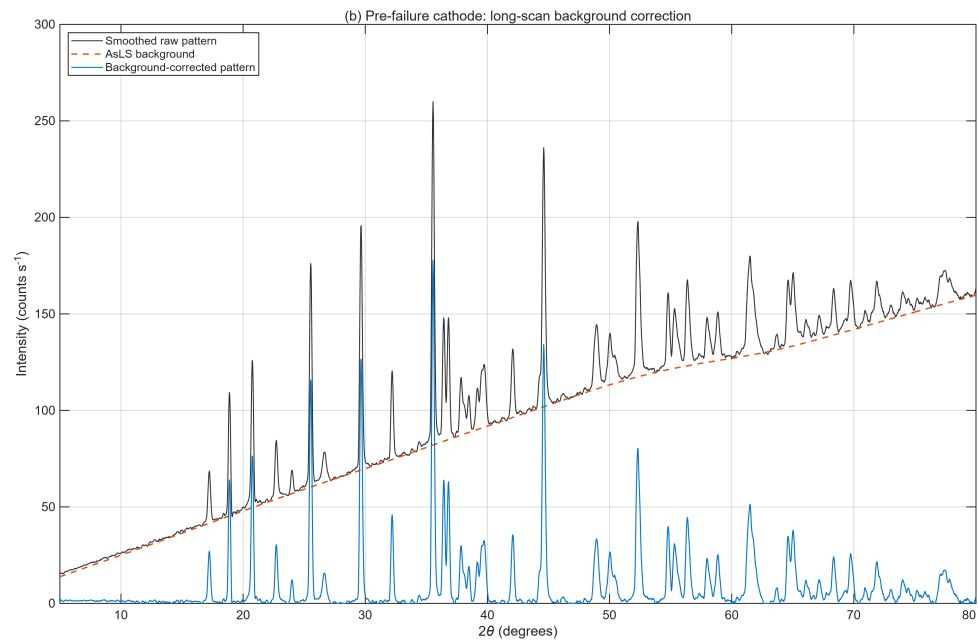


Figure 4.3: Asymmetric least-squares background fitting

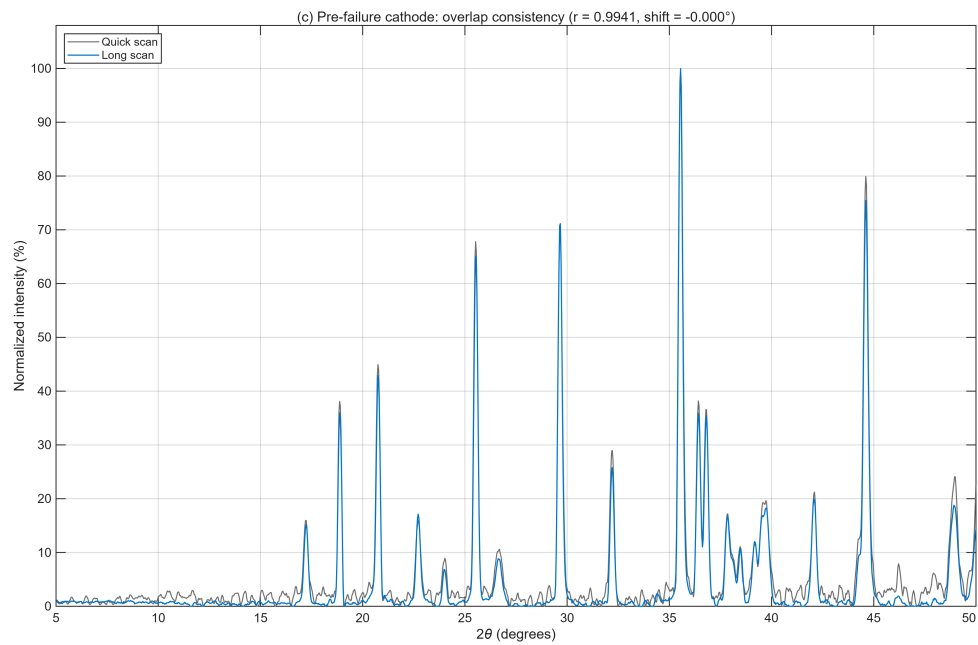


Figure 4.4: Comparison of Overlapping Normalized Curves

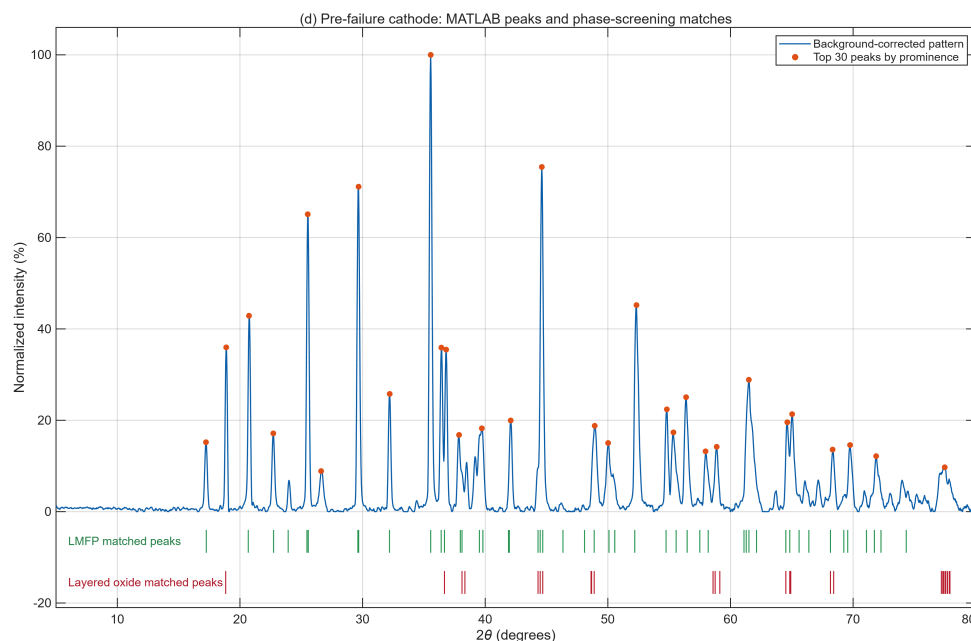


Figure 4.5: Preliminary peak matching

4.3. DSC Results

The DSC experiments use coin-type samples with LMFP/NMC cathodes. The reference sample does not have a cathode. The test temperature successively increases to 95, 100, 105, 110, 115, and 120 °C. It waits for 60 min at every stage, then it cools from 120 °C to approximately 29.3 °C. The differential heat flow curves of stepwise DSC show transient peaks at the onset of each step and upon entering the isothermal stage. The 95 °C isothermal stage shows a large negative shift, then it gradually recovers to nearly 5 mW. The 100–120 °C stage commonly shows a positive overshoot, then it decreases and gradually enters a stable plateau. After nearly 20–30 minutes at each isothermal stage, the temperature deviation and differential heat flow tend to become stable, which means it enters a relatively stable thermal equilibrium state. The average differential flow increases and follows the set temperature in the last 10 min of each isothermal stage.

This trend indicates that the differential thermal response between the sample and the reference increases with rising temperature. However, the coin-type structure does not have cathode material, this makes the sample and reference have a clear thermal capacity difference. The temperature control results show that the maximum temperature deviation at the 100–120 °C initial isothermal stage is nearly 1.5 °C, the minimum temperature deviation is nearly −0.1 °C. As the holding time extends, the temperature deviation is close to 0 °C.

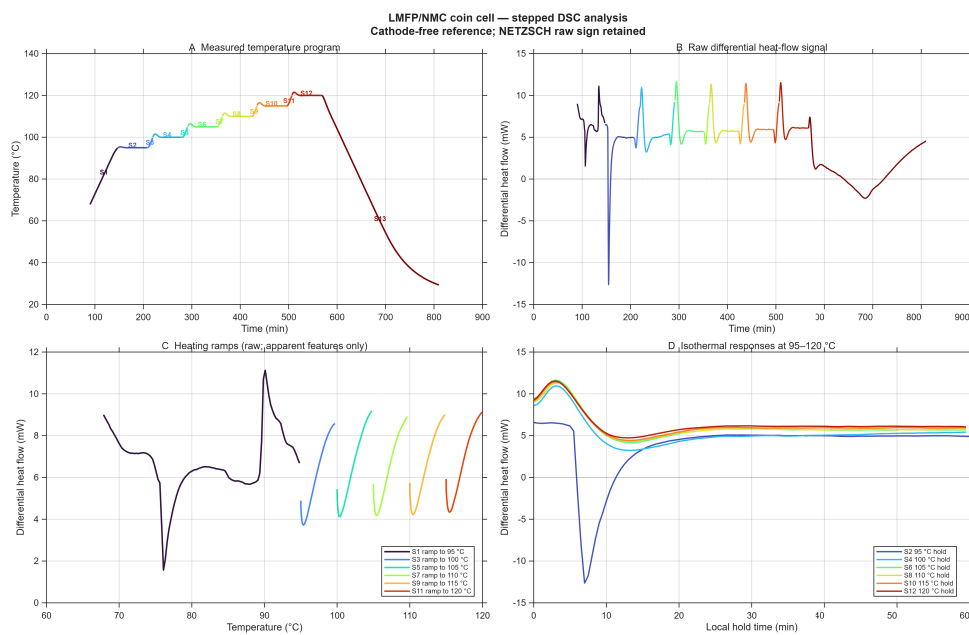


Figure 4.6: Measured temperature

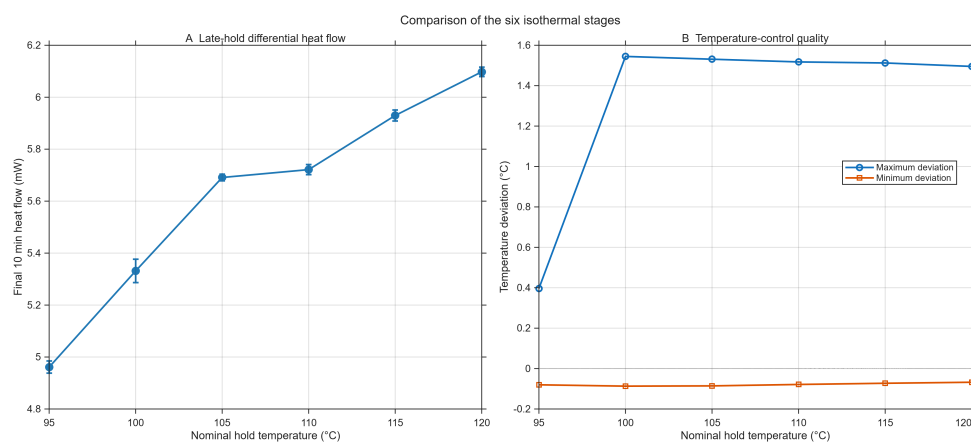


Figure 4.7: Initial heat flux response

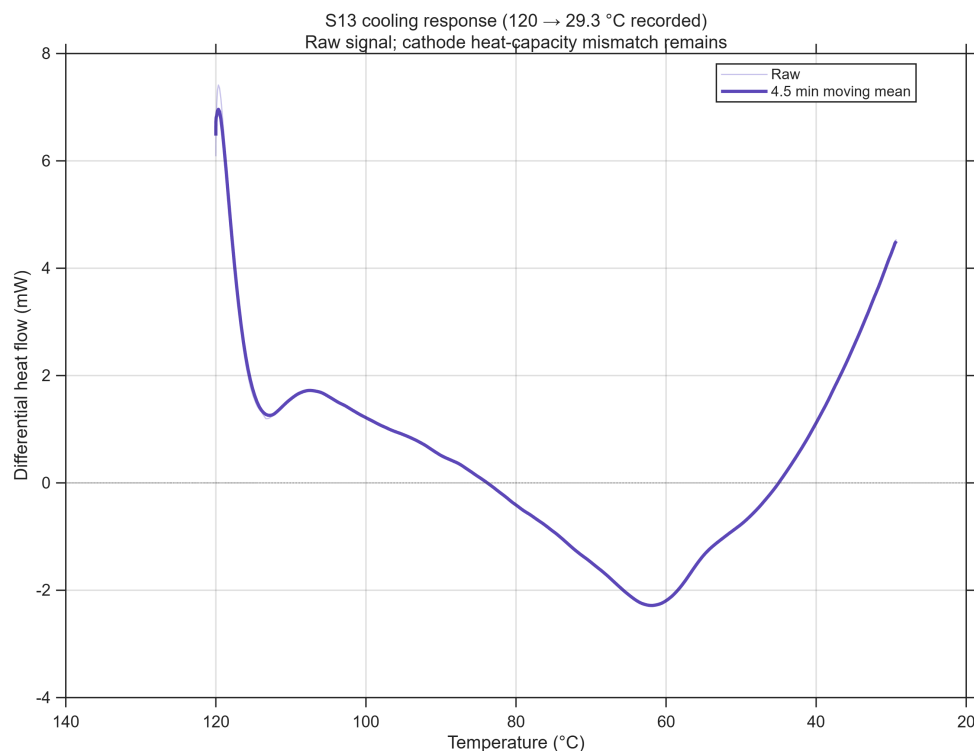


Figure 4.8: Differential heat flow change

When it cools from 120 °C to 29.3 °C, the differential heat flow did not return to a flat baseline which is near zero. It exhibits significant temperature dependence and crosses the zero point at 80–85 °C and 45 °C. This gradual curve did not exist in the cooling process of the continuous heating stage, which means the original DSC signal saved the difference in heat capacity between the sample and the reference.

4.4. TGA Results

The results of TGA show that the mass of the LMFP/MNC cathode sample decreases with the experiment time. The initial sample mass is nearly 8.47 mg, and it was nearly 8.23–8.24 mg at the end of the test. The total mass reduced by 0.23–0.24 mg, which is equivalent to nearly 2.7%–2.9% of the initial mass. During the initial heating and isothermal processes, the mass decreases gradually and constantly. The experiment did not observe an obvious single interval of rapid mass loss which belongs to the decomposition of the cathode active material. There are step-like changes visible around 36 min and 68 min, the latter change corresponds to the stage where the temperature program rises from approximately 106 °C to approximately 111 °C. The mass-temperature curve shows a nearly vertical mass change at the temperature of 105–111 °C, this is because the sample is maintained at a constant temperature. The highest temperature in the TGA test is 110 °C. The test does not cover large-scale melting of the separator, significant decomposition of the binder, or high-temperature instability of the layered oxide. This result mainly reflects the volatilization of the cathode and baseline mass stability when the temperature range is from low temperature to 110 °C before failure. It can not be used to determine the main decomposition temperature or total mass loss [17].

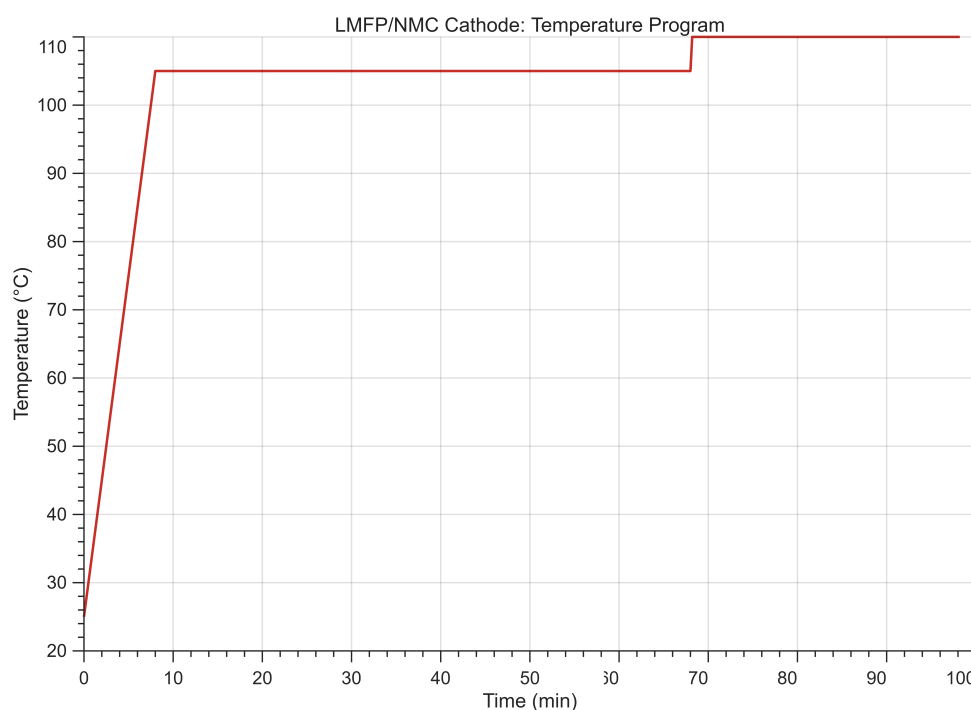


Figure 4.9: Experimental temperature program

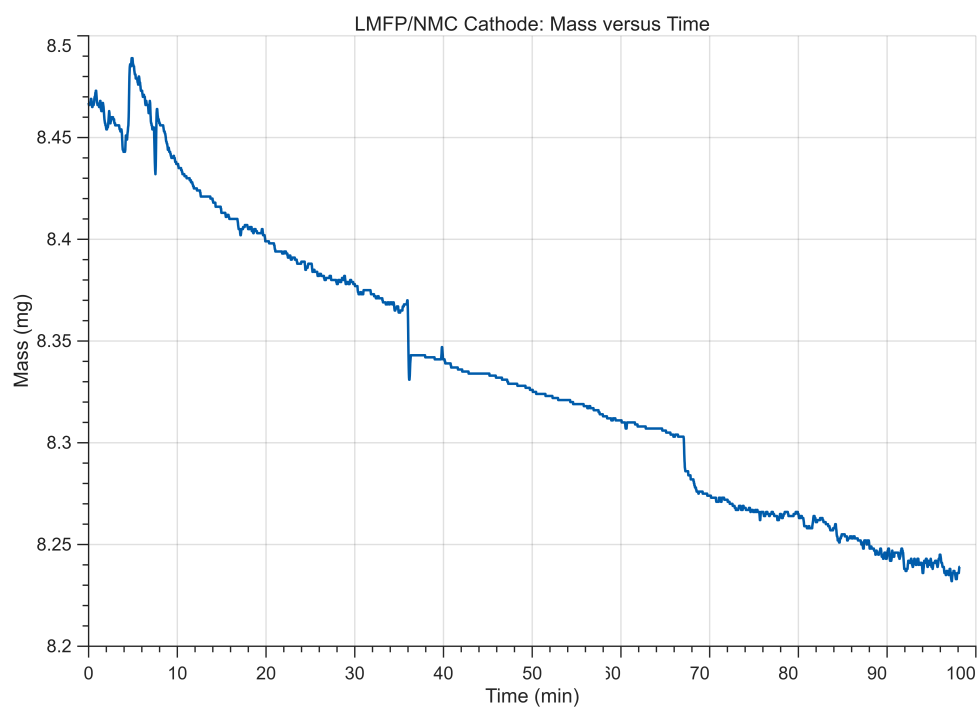


Figure 4.10: Change in mass over time

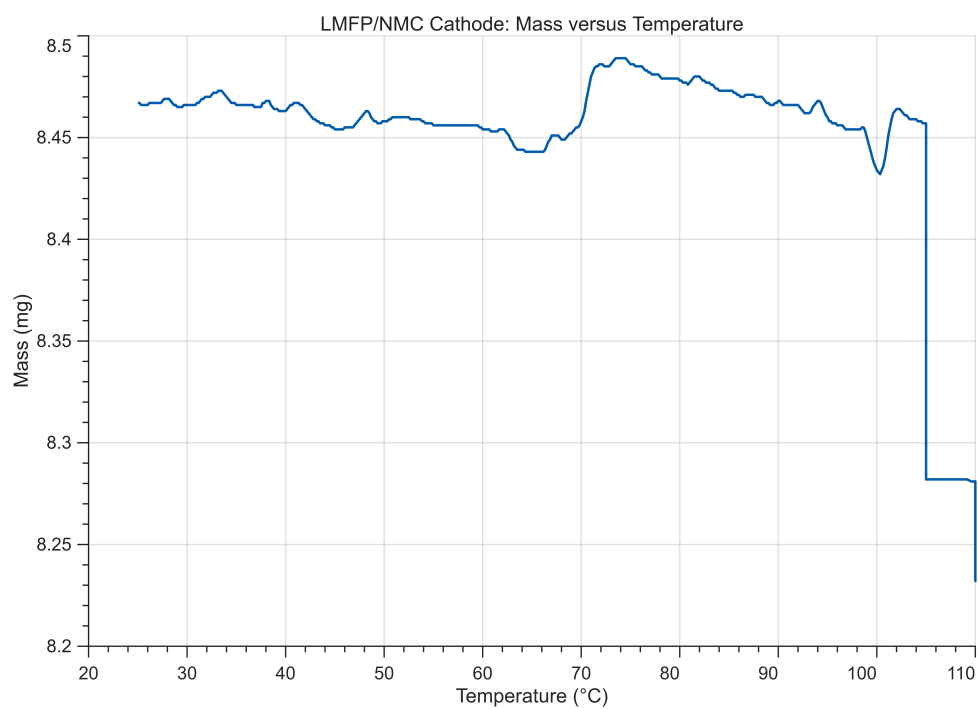


Figure 4.11: Variation of mass with temperature

4.5. Thermal response and voltage evolution in ARC Test

During drop 1, the terminal voltage drops from 3.98V to 2.45-2.50V. The instantaneous voltage drop is nearly 1.5V and contributes 37% – 38% of the pre-event voltage. There is a slight rebound and short-term fluctuation after the first drop and no clear temperature rise synchronous with the voltage drop is observed on the curve. Drop 1 is a discrete electrical event with stable, non-zero residual voltage, rather than a complete voltage collapse which is synchronous with the thermal runaway.

During drop 2, the voltage drops to 0.20V, this drop is 2.05V, and it contributes to 91% of the pre-event voltage. Drop 2 is bigger than drop 1 and makes the test voltage briefly approach zero. However, the peak value of recovered voltage after this drop is 3.2 V. This value is higher than the voltage before drop 2 which is 2.25 V and reached nearly 82% – 84% of the initial voltage of the ARC test. There is a 3 h duration between drop 2 and ADC overload, the self-heating rate is increasing continuously, but the voltage change process is non-monotonic. This results show that the measured terminal voltage can not be described by a single continuous discharge or a monotonic short-circuit process [19].

Event	Time (h)	Pre-event voltage (V)	Minimum/post-event voltage (V)	Main voltage response	Corresponding thermal state
First voltage drop (Drop 1)	≈ 19.6	≈ 3.98	≈ 2.45 – 2.50	Decrease of ≈ 1.5 V, followed by a stable non-zero voltage	≈ 110 °C; slow self-heating
Post-Drop 1 stabilization	≈ 19.7 – 23.6	≈ 2.50	≈ 2.25 – 2.40	Slight recovery followed by a gradual decrease	Gradually increasing self-heating
Second voltage drop (Drop 2)	≈ 23.6	≈ 2.25	≈ 0.20	Decrease of ≈ 2.05 V, corresponding to ≈ 91% of the pre-event voltage	≈ 125 – 130 °C; prior to the rapid temperature rise
Post-Drop 2 voltage recovery	≈ 23.7 – 26.2	≈ 0.20	Maximum of ≈ 3.4 – 3.5	Maximum recovery of ≈ 3.2 V with a non-monotonic response	Accelerating self-heating
ADC overload	≈ 26.6 – 26.7	≈ 3.4	Post-overload data not quantitatively valid	Termination of the valid voltage record	Approaching rapid temperature rise and thermal runaway

Table 4.1: Detailed stages data

The result of the ARC temperature shows that the sample entered thermal runaway quickly after

ADC overload, and the highest temperature was 450 °C; it deviated clearly from the average protection temperature. The anomalous recovery after the second drop occurs before the final thermal runaway, rather than during the cooling process.

The blue curve before the onset of overload is in the instrument's effective measurement range, it can be used for quantitative analysis of the two voltage drops and the latter voltage recovery processes. The gray curve which after the overload remains in 0–0.2 V, has been affected by range saturation and an anomalous measurement chain. It can only be used for indicate qualitative trends, but not for quantitative explanation of the real terminal voltage and it is not used to determine the time at which a short circuit occurs or when the voltage drops to zero.

Based on all the results of temperature, self-heating rate, and voltage, three continuous stages can be identified during the ARC process. The first is a low self-heating rate, the voltage drops slowly. Then is the anomalous voltage response stage before the onset of accelerated self-heating, its characteristic is two discrete voltage drops and a significant recovery following the second drop. The last is that the self-heating rate increases by orders of magnitude and enters thermal runaway. Both abnormal voltage drops occurred before the first rise in temperature. This indicates that the anomalous terminal voltage is an observable electrical response [18].

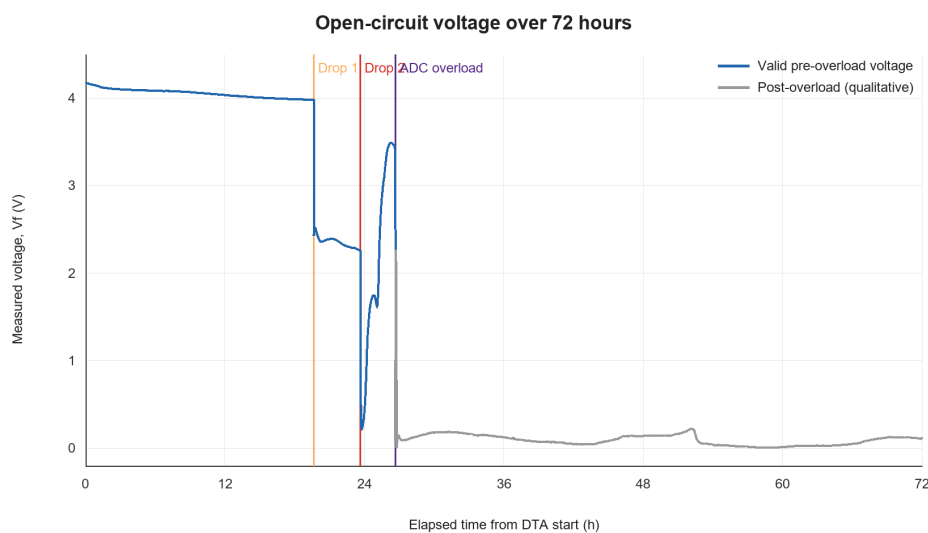


Figure 4.12: Open-circuit voltage

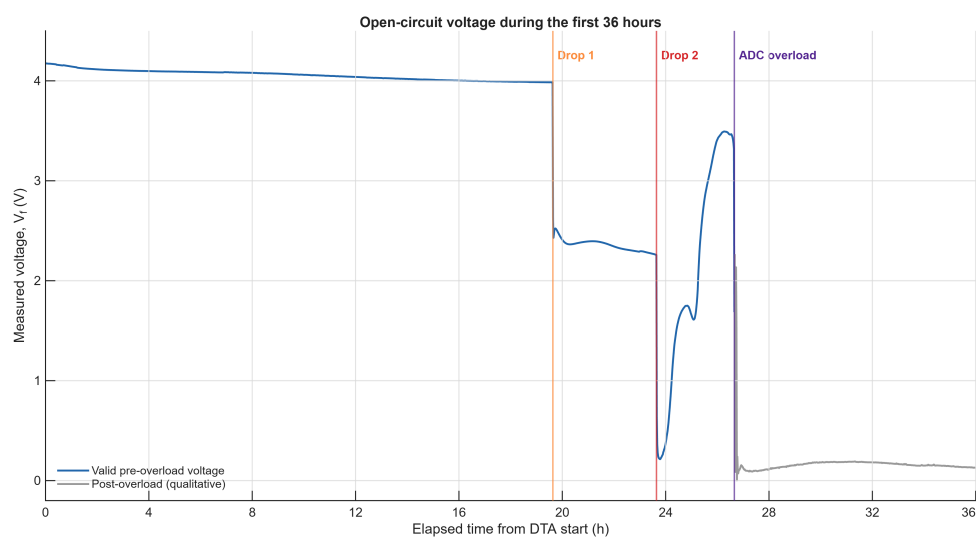


Figure 4.13: Enlarged view of a section

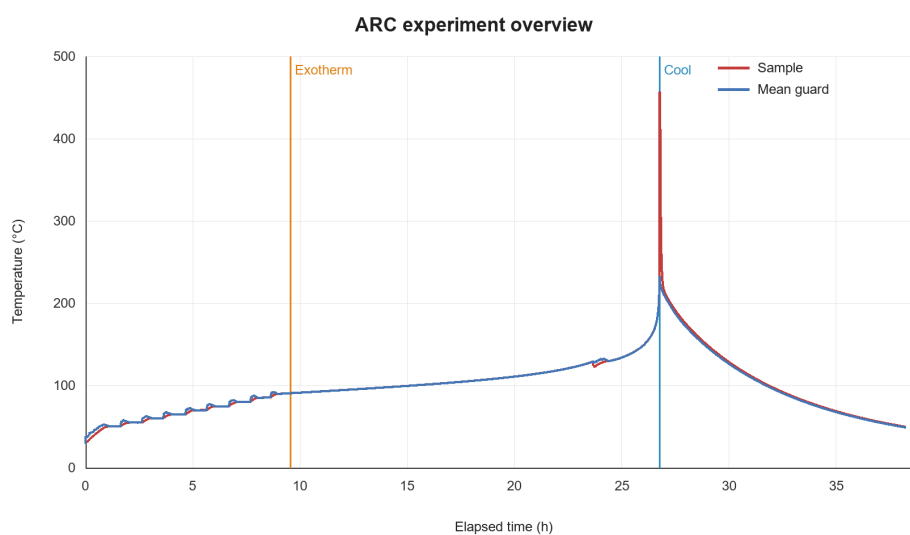


Figure 4.14: Variation of sample temperature and average guard temperature with test time

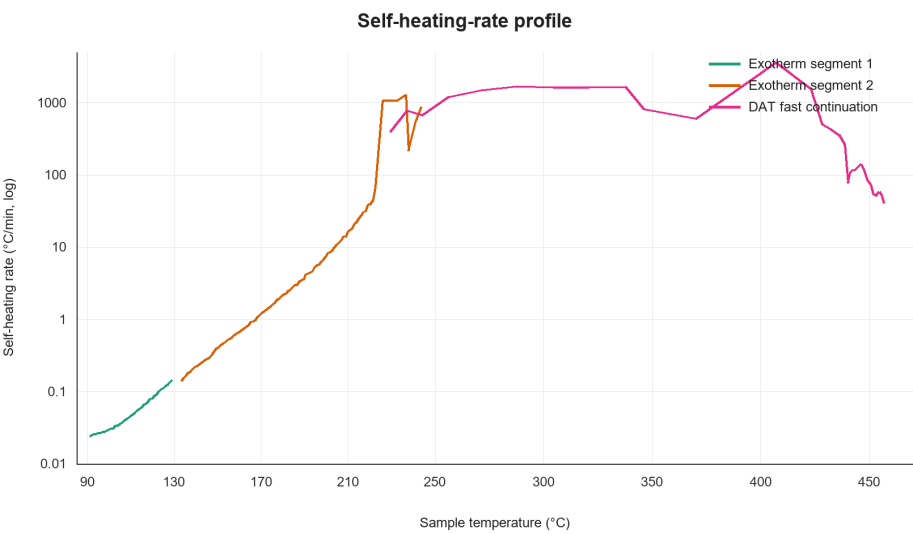


Figure 4.15: Self heating rate

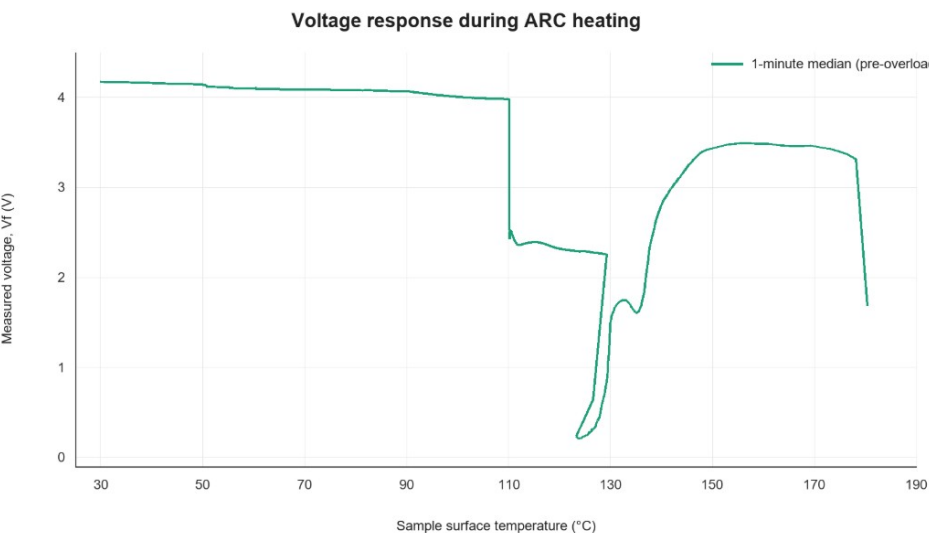


Figure 4.16: Voltages responses

Chapter 5

Conclusion

5.1. Battery structure and material state prior to failure

The CT axial cross-section shows a multi-layered structure in the region of the battery's cathode terminal. Many components that have shown in the results section have been observed, and they remain intact as a whole. Changes in the contact state of these components might affect the terminal voltage. The static CT results only show the initial potential relationship of these components.

The results of XRD show that the main diffraction peaks obtained from rapid scans and long-duration scans show good consistency regarding peak position and relative intensity. After the correction and phase screening, some diffraction peaks match the olivine-type LMFP phase, while the other peaks match the NMC-type of layered transition metal oxides. This proved the existence of two types of active materials in the cathode before failure. The XRD does not prove lattice changes, phase transitions, or structural collapse during ARC testing.

The results of DSC show that LMFP cathode samples exhibit temperature-dependent differential thermal responses in the 95–120 °C range. The transient peaks observed during the heating and isothermal stages show a correspondence with the actual temperature. The cooling process also maintains a non-zero differential heat flow and baseline crossing.

The results of TGA show overall mass decreases over time, there is no main mass loss range corresponding to the rapid decomposition of the primary cathode observed. Many reasons are related to this mass loss. The total mass loss can not be explained by structural degradation of LMFP or layered oxides.

5.2. Abnormal Voltage Response Prior to Thermal Runaway and Its Mechanistic Implications

5.2.1. Drop 1 Conclusion

- Before the first drop, the terminal voltage slowly drops from 4.17V to 3.98V. Then the voltage drops suddenly to nearly 2.45 – 2.50 V within a very small temperature range. At the same time range, the ARC temperature curves do not show a sharp temperature rise when the voltage drops. Therefore the drop 1 is a discrete electrical event occurring during the slow self-heating stage.
- Regarding the DSC results in the drop 1 temperature range, the DSC did not identify a distinct, sharp, and irreversible reaction peak corresponding to the sudden voltage drop. The weight loss of 2.7% – 2.9% obtained from the TGA results shows that the continuous thermal decomposition of the cathode is not enough to explain drop 1. This might also be related to complete destruction of the SEI and softening of the separator. The complete short circuit did not occur.
- The voltage drop and mass change occur earlier than the pressure relief and final thermal runaway, these signals can be early warning indicators. The low-temperature side reactions and volatilization processes create the conditions for heat generation, gas evolution, and a rise in pressure. The discontinuous electrical connection or change in structural state contributes to the anomalous voltage drop.

5.2.2. Drop 2 Conclusion

- During drop 2, the terminal voltage drops from 2.25V to 0.20V, and then recovers to 1.7-1.8V. After a brief drop, it rose rapidly again, reaching 3.4-3.5V. The 3.2V voltage recovery is obvious, but the rapid temperature rise did not coincide with the voltage minimum point.
- This temporal sequence does not match the sustained discharge and fast heat generation due to the permanent, hard internal short circuit. It is consistent with the voltage drop associated with transient electrical connection changes and the safety devices in the lit-

erature review. Drop 2 is more likely to show changes in the electrical connection on the cathode side, the top protective structure, or the effective connection between the terminal and cell core. The transient or localized internal short circuits can not be completely eliminated. However, the current data can not support take the near-zero voltage observed in drop 2 as the permanent internal short circuit.

- The reason of drop 2 might from a temporary loss of electrical connection between the end cap and the positive terminal tab, and related to the actions of CID. However, it is hard to distinguish between the activation of a safety device and internal short circuit based on the voltage drop across a cell equipped with CID.

5.3. Limitations

This research still has some limitations related to the experimental method and further improvement.

- The first limitation is that the CT, XRD, DSC, and TGA all belong to pre-failure and off-line characterization. There is no in-situ record of the dynamic evolution of the battery's internal structure and material state during the ARC process. This research can not directly prove the exact time of CID actions, disconnection of the cathode tab, separator shrinkage, and changes in the positive electrode crystal structure.
- The maximum temperature of DSC is nearly 120 °C, and the maximum temperature of TGA is nearly 111 °C; both of them only cover the initial self-heating and drop 1. The drop 2 occurs at nearly 125 – 130 °C, which is beyond the edge of the material testing range. This research can not distinguish the individual contributions of separator failure, severe electrolyte reactions, and high-temperature structural instability of the cathode.
- This research focuses on signal type and a 100% SOC cylindrical battery. The number of test samples and replications is not enough, thereby it can not evaluate the statistical distribution and repeatability of the two anomalous voltage drops. The current conclusions should be limited to the specific cell system, structure design, state of charge levels, and ARC test conditions. It should not be generalized to other types of batteries [20].

5.4. Future suggestions

The subsequent research should increase the number of ARC tests on the same type of battery and conduct statistical analysis on the time intervals of drop 1, drop 2, voltage recovery, and temperature. On this basis, it can improve the SOC, battery type, and cathode chemistry, thereby explaining the anomalous voltage responses to the initial state of energy, material system, and top structural design.

The material test also needs to be expanded to a broader temperature range and wider sample systems. The temperature range of DSC and TGA can cover at least 250 – 300 °C and test the cathode, anode, separator, and electrolyte. Meanwhile, the SEM, XPS, and high-temperature XRD can be applied to analyze the SEI, cathode surface, and collectors, thereby they can distinguish volatilization, active material phase change and metallic connection failures.

Once reproducible data have been generated, an early-warning criterion that combines voltage, surface temperature, self heating rate, and pressure signals can be established. The first non-zero voltage drop can serve as the signal of early anomalous activities, the second non-zero drop and recovery can act as the signal of escalation of failure state, the rapid self-heating rate can be used to identify the onset of thermal runaway. This kind of multi-signal, phased method can reduce false detection of internal short circuits caused by CID actions or transient electrical connections and enhance the stability of thermal runaway early-warning of cylindrical lithium-ion battery.

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Project Management Plan

Project Scope & General Objective: This project aims to investigate the failure modes and gaseous-solid emission characteristics of mainstream commercial lithium-ion batteries (specifically, BAK B18650-CX20M, LMFP/NMC hybrid cathode, 2 Ah capacity battery) under typical abuse conditions.

Work Breakdown Structure: [Phase 1: Preparation and Benchmark Characterization] → [Phase 2: ARC Abuse Testing] → [Phase 3: Emissions Acquisition and Microscopic Characterization] → [Phase 4: Data Integration and Writing]

Milestones:

Milestones	Core Deliverables	Timeline
M1: Preparation and Safety Assessment	Complete baseline XRD testing, battery OCV measurements, and laboratory safety training (COSHH/RA).	Week 1–3
M2: Accelerating Rate Calorimetry (ARC) Test Platform Setup	Complete continuity checks for the calorimeter gas-line piping, particulate filter, thermocouples, and voltage-monitoring leads.	Week 5
M3: High-Throughput Abuse Testing	Complete heat-wait-seek (HWS) adiabatic thermal runaway tests for all gradient samples and obtain the full real-time temperature-rise and pressure datasets.	Week 5–6
M4: Post-Failure Microstructural Characterization	Complete three-dimensional X-ray CT scans of the debris and compositional/phase analysis of gaseous and solid residues.	Week 7–8
M5: Final Deliverables	Complete cleaning of all datasets, construct the correlation matrix, and finalize and submit the thesis.	Week 9–10

Risk & Health Management:

Hazard	Controls
Explosion / Fire	Interlocked blast box; Class D extinguisher (Lith-X/L2); independent cut-off >600 °C.
Toxic / Corrosive Gas (HF)	BOFA LEV + lab exhaust;
Cuts / Burns	Insulate ≥1 terminal; keep blast box locked until <40 °C.

Project executor: Tianli Shi Supervisor: Alex Rettie

Experimental Programme Gantt Chart

Tianli Shi | Supervisor: Alex Rettie | June 1 - August 31, 2026

Task	Deliverable / Milestone	W1 Jun 1-7	W2 Jun 8-14	W3 Jun 15-21	W4 Jun 22-28	W5 Jun 29-Jul 5	W6 Jul 6-12	W7 Jul 13-19	W8 Jul 20-26	W9 Jul 27-Aug 2	W10 Aug 3-9	W11 Aug 10-16	W12 Aug 17-23	W13 Aug 24-31
Safety induction & COSHH/RA	M1 Safety approval	M1												
Literature review	D1 Review notes			D1										
Experimental protocol design	D2 Approved protocol		D2											
Supervisor meetings (Alex Rettie)	Progress notes													
Cell prep, OCV & baseline XRD	D3 Baseline dataset			D3										
ARC platform setup & checks	M2 ARC commissioned			M2										
HWS thermal-runaway testing	M3 Tests complete							M3						
Gas/particle collection & logging	D4 T/V/P + residue data							D4						
X-ray CT & residue analysis	M4 Characterisation									M4				
Data cleaning & correlation matrix	D5 Final dataset										D5			
Write Introduction & Methods	D6 Sections complete							D6						
Write Results & Discussion	D7 Full draft											D7		
Revision, checks & submission	M5 Final thesis													M5

■ Planning ■ Experiments ■ Analysis ■ Writing ■ Milestone ■ Deliverable



Battery Failure and Emissions

Document number:	APL-MRA-TS01
Location	APL
Supervisor	Alex Rettie
Version	1a
Date of assessment	10 April 2026
Overall risk after control measures	5- Very High

Personnel covered

Some processes may have reproductive toxicity which is potentially harmful to fertility and/or the unborn child. If, after reading this assessment, you are concerned about reproductive toxicity, please do not carry out this process, and contact the DSO to assess whether additional safety controls could be implemented.

Risk Summary – This will be a summary of the risks presented in this assessment and who is affected.

Summary of risks identified	1 Explosion/Fire	2 Exposure to hazardous materials	3 Electric shock	4 Burns	5 Impact/Personal Injury
	6 <u>Cuts/lacerations</u>	7	8 Click or tap here to enter text.	9 Click or tap here to enter text.	10 Click or tap here to enter text.

Persons at risk from the hazards associated with this activity/ task	Researchers ☒	Technical Staff ☒	Professional Services Staff ☒	Cleaners/ Maintenance staff/ Security staff ☐	External Contractors ☒	Members of the Public/ Visitors ☐
	Under-graduates ☐	Post-graduates ☒	Children / Young Persons ☐	Disabled Persons ☒	Pregnant/ Breast Feeding Persons ☒	Other Please specify here. ☐

Process overview/method statement:

Abuse testing of lithium-ion cells will be conducted to induce failure (thermal runaway) under controlled conditions. This includes thermal abuse, electrical abuse (overcharge, short circuit), and heating to destruction. [The purpose of this testing is to obtain electrochemical impedance measurements in the lead up to thermal runaway. Cell disassembly will be required prior to abusing cells to identify the electrodes via X-ray Diffraction \(XRD\) at the MFL.](#)

Testing will be carried out in enclosed systems (e.g. blast box) to contain ejecta, gases, and particulates. Emissions will be extracted via ventilation systems and may be collected for analysis.

Following testing, residues (particles and gases) will be analysed using:

- X-ray CT
- [XRD \(at MFL\)](#)

Carry out the ARC test in HEL. Heat-wait-see testing done on cylindrical cells at 100% SOC

Accelerated Rate Calorimetry (ARC) testing will be carried out using the HEL BTC130 calorimeter to investigate the thermal runaway behaviour of [BAK B18650-CX20M](#), LMFP/NMC blended 18650s [with 2 Ah capacity. The cells will be tested](#) at 100% SOC [\(4.2V\)](#). [The cells will not be cycled.](#)

Prior to testing, the cell will be visually inspected for damage and electrically characterised by measuring the open-circuit voltage. A K-type thermocouple will be attached to the surface of the cell using high-temperature tape. The cell will then be placed inside the stainless-steel ARC crucible and connected to the monitoring system for temperature and voltage measurements [\(charge and discharge at upper 4.2V, lower voltage 2.5V\)](#).

The crucible will be installed inside the HEL BTC130 calorimeter located within a ventilated enclosure. Gas outlet lines, particulate filters and gas collection vessels will be connected before testing. All safety interlocks and temperature limits will be checked prior to operation.

Heat-Wait-Search (HWS) testing will be performed. The calorimeter chamber will be heated incrementally to a predetermined temperature [\(Heat to 50 oC, wait half an hour, let cell self heat if temp rate exceeds 0.2 oC/min, otherwise increase temp by 5 oC\)](#) and then held for a specified waiting period. The system will monitor the cell for self-heating behaviour. If no self-heating is detected, the chamber temperature will be increased to the next step. If self-heating is detected, the calorimeter will track the temperature rise adiabatically until self-heating ceases or the cell enters thermal runaway.

During testing, cell temperature, chamber temperature, voltage and pressure data will be continuously recorded. Any gases released during failure will be directed through the extraction and collection system for subsequent analysis.–

The calorimeter chamber, crucible and associated fittings will be inspected after testing for signs of damage or contamination before further experiments are conducted.

The following criteria apply when deciding whether a cell may be safely removed from the ARC, or whether it must be destroyed:

- **Cells that have been instrumented to include an internal short-circuit device (ISC) and have been electrically-tested must be heated to destruction before removal from the ARC system.**
- **Cells that have been instrumented to include one or more internal thermocouples or strain gauges must be heated to destruction before removal from the ARC system if, at any point during testing, either of the following conditions are satisfied:**
 - **Their surface temperature exceeds the manufacturer's maximum temperature rating.**
 - **Either the internal (instrumented thermocouples) or external (cell surface and calorimeter guard heaters) temperature exceeds the maximum service temperature of any epoxy seals used in instrumenting the cell.**
- **If there is any uncertainty over whether all thermocouples have detached from the test cell, i.e. it is likely that the record of the cell temperature is inaccurate, then the cell must be heated to destruction before the ARC system may be opened, as described above.**

- If cell overcharged/discharged then it must be heated to destruction before removal from the ARC system.
- If a cell is at 0 V at the end of electrical testing and it has not heated at a rate of at least 100 °C/min, resulting in it's surface temperature reaching 250 °C, then it must be heated to destruction before removal from the ARC system.
- If a cell cannot be discharged to a safe level, i.e. the manufacturer's lower voltage cut-off, then it must be heated to destruction before removal from the ARC system.
- A commercial cell may be removed without heating to destruction if it has been discharged to a safe level, i.e. safe for shipping (30% SoC) or discharged no further than the manufacturer's recommended lower voltage cut-off (e.g. 2.5 V or 3 V for 0% SoC, **NOT** 0 V), and its surface temperature did not exceed the manufacturer's maximum temperature ratings during electrical testing.
- A cell that has been instrumented to include one or more internal thermocouples or strain gauges may be removed without heating to destruction if the three conditions below are **ALL** satisfied:
 - It has been discharged to a safe level, i.e. safe for shipping (30% SoC) or discharged no further than the manufacturer's recommended lower voltage cut-off (e.g. 2.5 V or 3 V for 0% SoC, NOT 0 V).
 - At no point during testing did its surface temperature exceed the manufacturer's maximum operating temperature.
 - At no point during testing did either the internal (instrumented thermocouples) or external (cell surface and calorimeter guard heaters) temperature exceed the maximum service temperature of any epoxy seals used in instrumenting the cell.
 - A cell may be removed without heating to destruction if it is at 0 V and it heated at a rate of at least 100 °C/min, resulting in its surface temperature reaching at least 250 °C during testing. The cell should be treated as having failed successfully, so cleanup/removal should be conducting according to the hazard mitigation strategies outlined in this document.

  Toxic ☒ Oral toxicity ☒ Dermal toxicity ☒ Inhalation Toxicity ☒ Aspiration Toxicity	  Corrosive ☒ Skin corrosion / burns ☒ Eye damage ☒ Corrosive to metals	  Explosive ☒ Self-reactive ☒ Organic peroxides
  Serious long term health hazard ☒ Respiratory Sensitiser ☒ Carcinogen ☒ Mutagen ☒ Reproductive Toxicity ☒ Target Organ Toxicity	  Flammable ☒ Pyrophoric ☒ Self-heating ☒ Emits flammable gases ☒ Self-reactive ☒ Organic peroxides ☐ Aerosols	  Gases under pressure ☒ Toxic gas ☒ Asphyxiant ☐ Inert/non-haz.
  Harmful ☒ Irritant ☒ Skin sensitizer ☒ Serious eye irritation ☐ Narcotic effect	  Oxidising	  Dangerous to the environment

Use table 1 to complete the list of chemicals used (individual component substances, not combinations e.g. commercial/brand products/names) and their hazards

(A) Health Hazard	(B) Dustiness or *Volatility	(C) **Quantity	Score
Low Hazard Statements: H303; 305 H313; 315; 316 H320 H336 EUH066	Low Solids: Pellet-like solids that do not break up. Little or no dust observed during use. Solids forming large crystals Liquids: Boiling point > 150°C	Small <1g (ml)	1
Medium Hazard Statements: H301; H302 H311; H312; H314; H317 H318; H319 H332; H335 H371; H372; H373	Medium Solids: Smaller crystalline or granular solids. Minimal dust, or if any dust is seen it settles out quickly. Liquids: Boiling point between 50 and 150°C	Medium 1 to 100g (ml)	2
High Hazard Statements: H300; H304 H310 H330; H331; H334 H340; H341 H350; H351 H360; H361; H362 H370 EUH070	High Solids: Fine, light powders. Dust can be seen during use and possibly remains airborne for several minutes. Liquids: Boiling point < 50°C	Large > 100g (ml)	3

Multiply (A)x(B)x(C) to estimate overall risk level: ≤ 7 Low; 8-11 Medium; ≥ 12 High

*if a chemical has more than one risk phrase, use the one(s) with the highest health hazard score¶

**When stating quantity, this should consider the quantity in the stock bottle as well as the quantity of the aliquots, since loss of containment from the entire stock bottle whilst removing aliquots may represent the greatest risk.¶

Use the overall risk level (i.e., the worst) and determine as follows to take forward to risk assessment form. Tick below the relevant number:

Table 1				
Chemical	Health Hazard Score (A)	Dustiness Volatility Score (B)	Quantity Score (C)	Overall Risk Level (AxBxC)
Nickel oxide	<u>3</u>	<u>3</u>	<u>2</u>	<u>18</u>
Cobalt oxide	<u>3</u>	<u>3</u>	<u>2</u>	<u>18</u>
Manganese oxide	<u>2</u>	<u>3</u>	<u>2</u>	<u>12</u>
Hydrogen (H₂)	<u>1</u>	<u>2</u>	<u>2</u>	<u>6</u>
Hydrogen fluoride (HF)	<u>3</u>	<u>3</u>	<u>2</u>	<u>18</u>

Very low	Low	Moderate	High	Very high
1,2	3,4	6,8	9,12	18,27
1	2	3	4	5
☐	☐	☐	☐	☒

Electrical equipment list:

THT or HEL BTC-130 ARCs for abuse

[Computer](#)

[Multimeter](#)

[Gamry/Ivium potentiostat](#)

Kaercher NT 75-1 Me Ec H Z22 vacuum cleaner

Tool list:

Electrical tape

Tweezers

Cell Holders

Thermocouples

Relevant COSHH, risk assessments, and SDS:

APL-CA-006 Fire extinguisher powder for lithium-ion cells

APL-CA-059 Battery Failure Products

APL-RA-098 Destructive calorimetry of Li-ion, Li-S, & Na-ion cells (≤ 5 Ah) for pressure measurements, and gas and particulate collection in the HEL BTC130

[APL-LRA-001 General Lab Risk Assessment](#)

[APL-RA-057 Accelerated rate calorimetry and operational abuse with the THT ARC system](#)

[APL-SPEC-017-BAK B18650CX-20M](#)

Control measures: list below control measures e.g., extraction, ventilation, supervision, include additional controls for vulnerable persons where necessary **Tick boxes**

Glove box (argon filled, or negative pressure isolator)	☐	Fume hood (standard lab ducted, built-in)	☐	LEV (ductless fume hood, Nederman or other system)	☒
Ventilation (air conditioning)	☐	Bench top	☒	Other	☒
Other/notes (please specify details of control system):	ARC equipment contained in aluminium enclosure around extraction canopy. Access to both areas restricted. BOFA filtration unit attached to the containment chambers.				

Personal Protective Equipment identify type and specification, please see here for example Tick boxes					
	☐			☒	Helmet as part of respirator
	☒	Powered/supplied air respirator; EN14387-compliant		☒	EN 166 FT
	☒	Neoprene EN ISO 374-1/Type B (30 minutes) (K[inorganic base]P[eroxide]T[aldehyde]) (green) gloves		☒	CE Cat I, type 5/6
	☒	Safety boots		☒	Galoshes and safety coveralls ASTM T903

Is health surveillance or monitoring required? (health surveillance is absolutely necessary for any loose carcinogenic or respiratory sensitising substances, and may also be required for vulnerable persons e.g., pregnant/young workers those with asthma, dermatitis etc) Tick boxes	Yes ☐	No ☒
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First aid measures – check SDS for information
General advice: Consult a physician. Show this COSHH form and associated MSDS for failed cell to the doctor in attendance. If inhaled: Move person into fresh air. Consult a physician immediately. In case of skin contact: Wash off with soap and plenty of water. If irritation occurs, consult a physician immediately. In case of eye contact: Flush eyes with water, including under eyelids, for at least 30 minutes. Take care not to rinse contaminated water into other eye or elsewhere that it may contaminate. Saline may also be used. Consult an eye specialist/transport to emergency care immediately. If swallowed: Rinse mouth thoroughly with water. NEVER give anything by mouth if victim is rapidly losing consciousness. DO NOT INDUCE VOMITING. If vomiting occurs naturally, have victim lean forward to reduce risk of aspiration. Rinse mouth thoroughly with water again. Transport to emergency care/contact local poison control centre immediately. LD50: Primary substances that a user is most likely to come into contact with are graphite and NiO: Graphite acute toxicity LD₅₀: >2000 mg/kg NiO LD₅₀ acute toxicity: >5000 mg/kg

Fire - identify appropriate fire extinguishers here *Tick boxes*

Powder D for lithium fires Lith-X / L2	Powder D for lithium fires Lith-X / L2 in BAGS	Powder D for metal fires – NOT LITHIUM M28 (NaCl)	Sand	Water mist	Water	Other
☒	☒	☐	☐	☐	☐	☐

During combustion substances may give rise to harmful vapours/gases etc please detail below

NA - combustion is the aim of the process and substances are listed above.

Storage

Cells are stored in the battery cabinets at the safe transport voltage (typically 30% SOC) prior to experimentation.

Insulate the Terminals: Prevent accidental external short circuits.

Temperature Control: Store the cells in a dedicated, climate-controlled room or refrigerator maintained at a stable **15°C to 25°C**.

Each stored cell/container will be clearly labelled with the cell type/model, lithium-ion battery hazard warning, storage date, responsible person, and storage voltage or SOC.

Gases are given off as cell vents or undergoes combustion – fed into BOFA LEV which scrubs heavy species (not CO₂, CO, hydrogen, lighter hydrocarbons, NOx), and cleaned air vented through the labs extraction system (only non-toxic gases and below-WEL CO, NOx should remain).

If necessary, gases may be stored in sealed containers in ductless fume hood or test chamber (ARC) with extraction running.

Solid decomposition products left to cool and settle at the bottom of the main chamber/calorimeter crucible. Before crucible is opened, main chamber is vacuum cleaned using the **Kaercher NT 75-1 Me Ec H Z22 vacuum cleaner** – cleaned and maintained.

At the end of its service life, the vacuum cleaner will be disposed of through the University's approved recycling route. After opening the crucible, all large fragments of interest are removed and placed into a correctly labelled plastic sample bag. Bag is stored in an ammunition box in a designated storage cupboard. All material has fully decomposed and poses no risk if properly sealed in bag.

Disposal of waste substances and containers please indicate below *Tick boxes*

Battery disposal	Hazardous Waste	General waste	Return to supplier	Other (specify)
☒	☒	☐	☐	☐

Other:

Hazard	Associated risks	Severity	Probability	Control measures Preventative and mitigative	Probability after control	Score
Removing failed cell from blast box	Exposure to hazardous materials, carcinogens, allergen/sensitizer by respiratory route. Burns from heated surfaces during test. Fire uncontained ignition of the flammable materials within the cell, or an unexpected cell failure. Personal injury from contact with sharps associated with failed cell	5	4	For more information read <i>Waste, storage, decontamination required at end of experiment or after carrying out this process.</i> Engineering controls: - Blast box door will not open until internal temperature has dropped below 40°C. BOFA and extraction canopy are in place to remove harmful particulates and gasses. BOFA unit must be running at full speed before clean-up can begin. Enclosure walls are in place to prevent release of material and prevent untrained users from approaching the calorimeters. Procedural controls: Sample must be confirmed as destroyed or fully discharged before opening the blast box. See the points highlighted in yellow in this RA for more detail. Do not put head inside the blast box. - Personal protective equipment: - Two layers of neoprene gloves (EN 374). - Cut-resistant gloves underneath neoprene gloves if necessary. - Solvent mask/respirator (3M 4277+ or 4279) that has been fit-tested OR powered/supplied air respirator; EN14387-compliant. - Face shield. - Protective coveralls.	2	8
Running a failure test	Exposure to hazardous materials ,	4	4	Engineering controls: Blast box door must be closed and interlocked before beginning test.	2	8

	carcinogens, allergen/sensitizer by respiratory route. Burns from heated surfaces during test. Fire uncontained ignition of the flammable materials within the cell			• BOFA and extraction canopy are in place to remove harmful particulates and gasses. • Enclosure walls are in place to prevent release of material and prevent untrained users from approaching the calorimeters. • The blast box door cannot be opened while the interior thermocouples read above 40°C. • If the software or system crashes, such that the heater control malfunctions, the heater power supply has a separate master cut-off that will turn off the heaters if the calorimeter temperature exceeds 600 °C. Procedural controls: • Abuse testing protocols should, where possible, be configured such that cell failure occurs when there are minimal users in the lab, i.e. after 7pm on weekdays, or at the weekend, to reduce the risk of exposure to lab users should any vented gases leak from the system. ◦ With appropriate dust/gas leakage mitigation strategies in place, it should be safe to run tests when users are present if necessary (e.g. if user presence in the lab is required to adjust monitoring equipment). Personal protective equipment: • Two layers of neoprene gloves (EN 374). • Solvent mask/respirator (3M 4277+ or 4279) that has been fit-tested OR powered/supplied air respirator; EN14387-compliant • Face shield • Protective overall Gloves should not be worn when using clean computer equipment		
Handling and transport of charged cells	Fire/burns from external short	4	4	Engineering controls: • Ensure at least one terminal on the cello is covered	2	8

	Pinch points from crocodile clips/cell tabs Crushing or impact from placement or dropping of heavy calorimeter lid Exposure to hazardous electrolyte caused by a leaking cells or venting (leading to eye and skin damage, and respiratory irritation)			with insulating tape. Do not allow both terminals to be exposed at the same time. - Electrical wires and connections must be sheathed in thermally-stable materials (e.g. Vidadflex glass/resin sheathing, ceramic beads, thermal tape) to reduce the risk of electrical contact between the cell terminals. Thermocouples must be sheathed in thermally-stable materials in order to reduce the risk of electrical contact between the cell casing and the metal parts of the testing unit. Procedural controls: - All users will be trained by a competent person before carrying out this process unsupervised. - Extreme care must be taken when handling cell to set up test. Do not place the cell directly on any metallic surfaces. Instead, use a non-conductive, heatproof mat during setup and ensure that the cell is stable if put down. L2 powder must be readily available to the user to drop on the test cell in case cell shorting occurs and it is not possible to place the cell within the blast box. Check cell voltage before beginning. Cells should only be handled at 30% SOC. Visually inspect cells before transfer to the calorimeter. Check for possible leaks, dents, deep scratches etc. If the cell is damaged do not use it. Store it in a battery safe and contact the lab manager or DSO to arrange disposal. - Only potentiostats in which the working electrode is NOT connected to ground should be used.		
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				• All electrical attachments to the cell must be checked for non-continuity before attaching to the cell, i.e. between the positive and negative terminal leads using multimeter – an open circuit (very high resistance, > 1 GΩ) is essential before test setup may begin. • The blast box door should always be free of obstruction, e.g. wires, thermocouples, so that it is always ready to close easily. • If the gas collection canister is to be used, then inline filters MUST be used to prevent particles from entering the collection line. • For gas collection, a water trap must be employed in order to allow gases to vent to the BOFA without passing through water. • The calorimeter an/or ammo box (if being used for testing) MUST be insulated as far as possible with e.g. superwool in order to help catch particles and contain the cell failure. We have found this method, of improved insulation, to be effective in reducing lab exposure to particles. • Place additional insulating panels (e.g. superwool, heatproof mats) between the calorimeter and blast box door, inside the blast box. We have found this to be an effective way of mitigating particle leakage into the lab. Personal protective equipment: • Two layers of neoprene gloves (EN 374). • Solvent mask/respirator (3M 4277+ or 4279) that has been fit-tested OR powered/supplied air respirator; EN14387-compliant • Face shield		
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				Protective coveralls		
Use of electricity, AC and DC, to power equipment	Death from electric shock via contacting live voltage Burns from contact with live current Fire from equipment fault or water ingress	5	3	Engineering controls: - Equipment designed and manufactured for purpose. Procedural controls: - Equipment is used as received from the supplier and is CE/UKCA marked. - Equipment is PAT tested as required under UCL policy. - No unauthorised alterations or repairs to be made to equipment. - Equipment used is compatible with lab supply voltage. - Equipment is maintained and checked regularly for damage/deterioration. - Damaged equipment is reported and taken out of use. - Experimental card placed on setup to indicate safe switch-off procedure in the event of an emergency. - In the case of electrocution of fire, switch off equipment at the wall/mains supply. Water mist fire extinguishers available. - Only use fire extinguishers if comfortable doing so. Personal protective equipment: - N/A – high voltages will not be used in this process and electrical work/repair/maintenance of powered devices will not be carried out.	1	5
Setup, use and dismantling of equipment – placing and carrying loads before experiment begins, or during if	Impact injury from dropping objects Muscular strain/injury from	3	4	Engineering controls: Lifting equipment will be used where necessary to move or place large object, e.g., trolley, pallet truck. Ask for assistance from estates for handling of any larger items, e.g., that might require a forklift or other equipment which requires training.	2	6

components need to be moved	improper manual handling Trip/fall from obstructive object placement			Procedural controls: • Ask for assistance with any loads that are challenging to carry, as necessary. • Practise proper lifting technique. • Break loads into smaller consignments if possible. • Careful management of equipment placement and lead connections to avoid trip hazards and blocking of travel or fire escape routes (good housekeeping). • Placement of electrical leads to avoid trapping in doors/furniture (good housekeeping). • Materials must also be placed for easy access in an emergency, e.g., PPE, tools. • Lab space checked regularly for damage/deterioration. ◦ Any damage is reported, and the area cordoned off/signage put up as necessary to notify users. Personal protective equipment: • Safety shoes and packing gloves for manual handling of large items.		
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